DataMites Global Training Institute

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PROJECT_ID: PTID-CDA-MAR-26-1362

PROJECT_NAME: PRDA – 01 Profit Analysis

DataMites Global Training Institute

PRDA – 01 Profit Analysis

Create new Connection

Connection of MySQL

- **Host:** **********
- **Name:** project_profit_analysis
- **Table Name:** startup
- password =**********
- database = "project_profit_analysis"

Multiple Linear Regression & Predictive Modeling

Project Objectives

- ✓ **Goal:** Identify how R&D, Administration, and Marketing spend affect company Profit.
- ✓ **Key Question:** Which investment area drives the most growth?
- ✓ **Tool:** MySQL for data extraction; Statistical Regression for modeling.

Regression Model Overview

- The Equation:
- $Profit = \beta_0 + \beta_1(R\&D) + \beta_2(Admin) + \beta_3(Marketing)$
- Dependent Variable (Y)
- Profit Independent Variables (X):
  - X1: R&D Spend
  - X2:Administration Spend
  - X3: Marketing Spend

Key Insights (Coefficient Analysis)

- ✓ R&D Spend: Strongest positive impact (b1 ≈0.85$). For every $1 spent on R&D, profit increases significantly.
- ✓ Marketing Spend: Moderate positive impact. Good for growth but less efficient than R&D.
- ✓ Administration: Weakest correlation. This is often an "overhead" cost and doesn't directly boost profit.

Regression Analysis(Task 2) by Using Excel

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.975062
R Square	0.950746
Adjusted R	0.947534
Standard E	9232.335
Observatic	50

ANOVA

	df	*SS*	*MS*	*F*	*ignificance F*
Regressior	3	7.57E+10	2.52E+10	295.9781	4.53E-30
Residual	46	3.92E+09	85236007		
Total	49	7.96E+10			

	Coefficients	*andard Erro*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*ower 95.0%*	*Upper 95.0%*
Intercept	50122.19	6572.353	7.626218	1.06E-09	36892.73	63351.65	36892.73	63351.65
RD_Spend	0.805715	0.045147	17.84637	2.63E-22	0.714838	0.896592	0.714838	0.896592
Administra	-0.02682	0.051029	-0.52551	0.601755	-0.12953	0.0759	-0.12953	0.0759
Marketing_	0.027228	0.016451	1.655077	0.104717	-0.00589	0.060343	-0.00589	0.060343

Cont…

Steps:									
	2)	independent variable(X) input X range: R&D(research and Development), Administration, and Marketing							
		Dependent Variable: Profit Column(including the header)							
	3)	the output							
		Variable		coeffficients					
		intercept		50122.19					
		R&D Spend		0.805715					
		Administration		-0.02682					
		Marketing Spend		0.027228					
	4)								
		predicting for inputs	Scenario A:		scenario B:				
		RD	21892.92		23940.93				
		Admin	81910.77		96489.63				
		Marketing	164270.7		137001.1				
Furmulas		Calculation= Intercept+(β1*RD)+(β2*Admin)+(β3*Marketing)					where **β = coefficients**		
scenario	i)	70037.90477							
scenario	II)	70554.57256							

Description For the Above Excel results

A. Regression Statistics

This section evaluates the overall fit and strength of the model:

- Multiple R (0.975): Shows a very strong positive correlation between the spending features (R&D, Admin, Marketing) and Profit.
- R Square (0.9507): Approximately 95% of the variance in Profit can be explained by the money spent on R&D, Administration, and Marketing.

Cont....

- **Adjusted R Square (0.9475): This is the more reliable version of R Square as it accounts for the number of variables used. It confirms the model is highly accurate.**
- **Observations (50): Confirms the analysis was conducted on the 50 startups mentioned in the project brief.**

2. ANOVA

This table determines if the model is statistically significant

- **Significance F (4.53E-30): This value is extremely close to zero. Since it is much lower than 0.05, it means the model is highly significant and the relationship between spending and profit is not due to random chance.**

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Variable Analysis (Coefficients & Significance)

- **The *P*-values indicate whether a variable has a statistically significant impact on profit (typically a *P*-value < 0.05 is significant).**

Key Points

1. R&D – is the most important predictor for these startups.it is the only spending category that is statically significant at the 95% confidence level.
2. State doesn't matter much when compared to R&D.
3. Marketing & Admin: while marketing shows a slight positive trends, its impact is secondary to R&D. Administration costs shows a slight negative coefficient but are statistically irrelevant to profit outcomes.

Regression Analysis Using Python

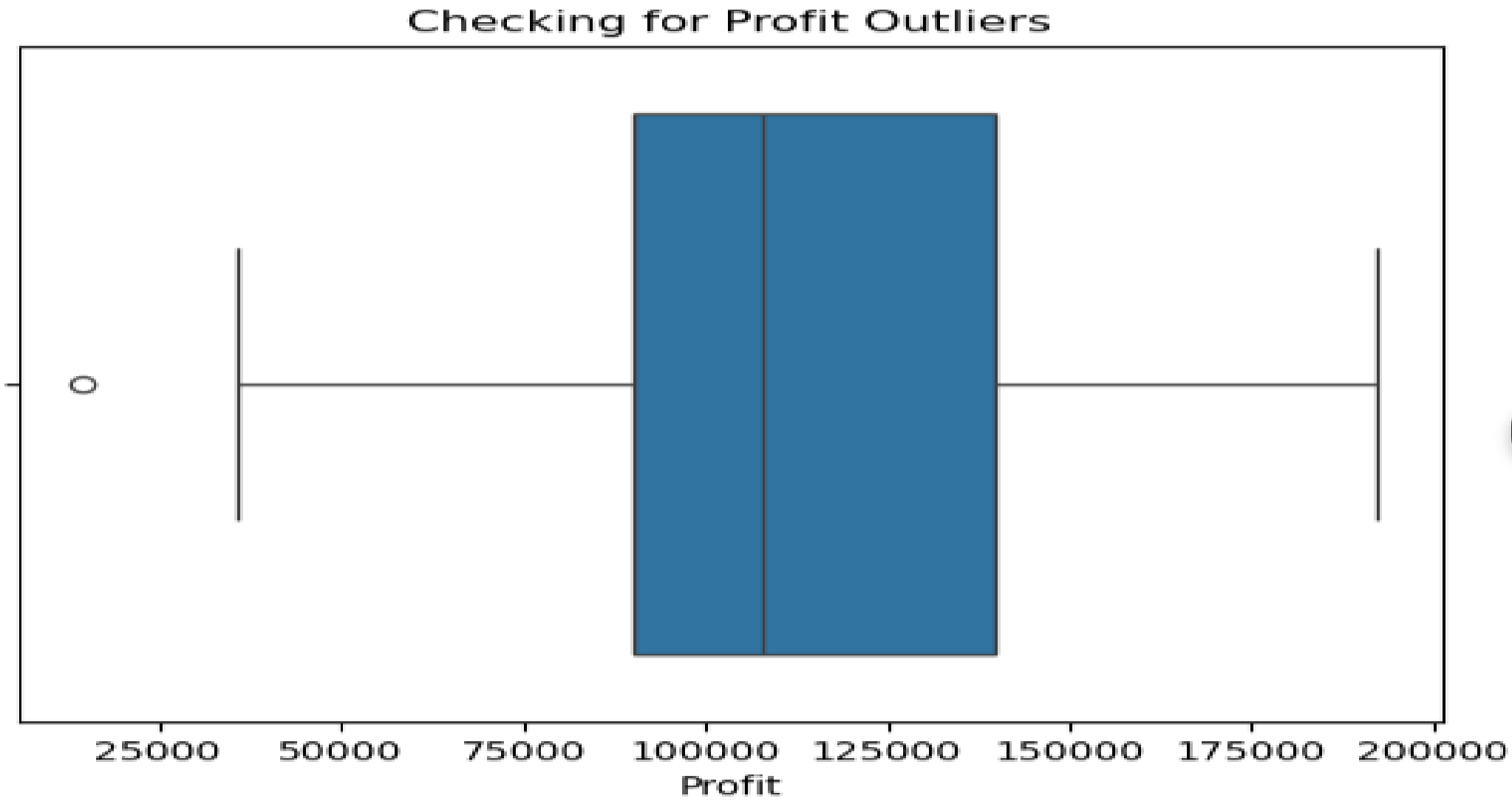
```
Features used: ['RD_Spend', 'Administration', 'Marketing_Spend']
Target used: Profit

--- REGRESSION ANALYSIS RESULTS ---
                  Coefficient
RD_Spend             0.805715
Administration      -0.026816
Marketing_Spend      0.027228

Base Profit (Intercept): $50,122.19

--- PREDICTIONS ---
Scenario 1: Predicted Profit = $70,037.90
Scenario 2: Predicted Profit = $70,554.57
```

Boxplot


Checking for Profit Outliers
0
25000
50000
75000
100000
125000
150000
175000
200000
Profit

. The Components of the Boxplot

- The Blue Box: This represents the "Middle 50%" of your startup profits. Most of your companies are earning between roughly 90,000 and 140,000.
- The Center Line: This is the Median (the exact middle profit value), which looks to be around 108,000.
- The "Whiskers" (Horizontal Lines): These show the spread of the data. The right whisker shows your highest-earning normal startups (up to nearly 200,000), and the left whisker shows the lowest normal profits (down to about 35,000).

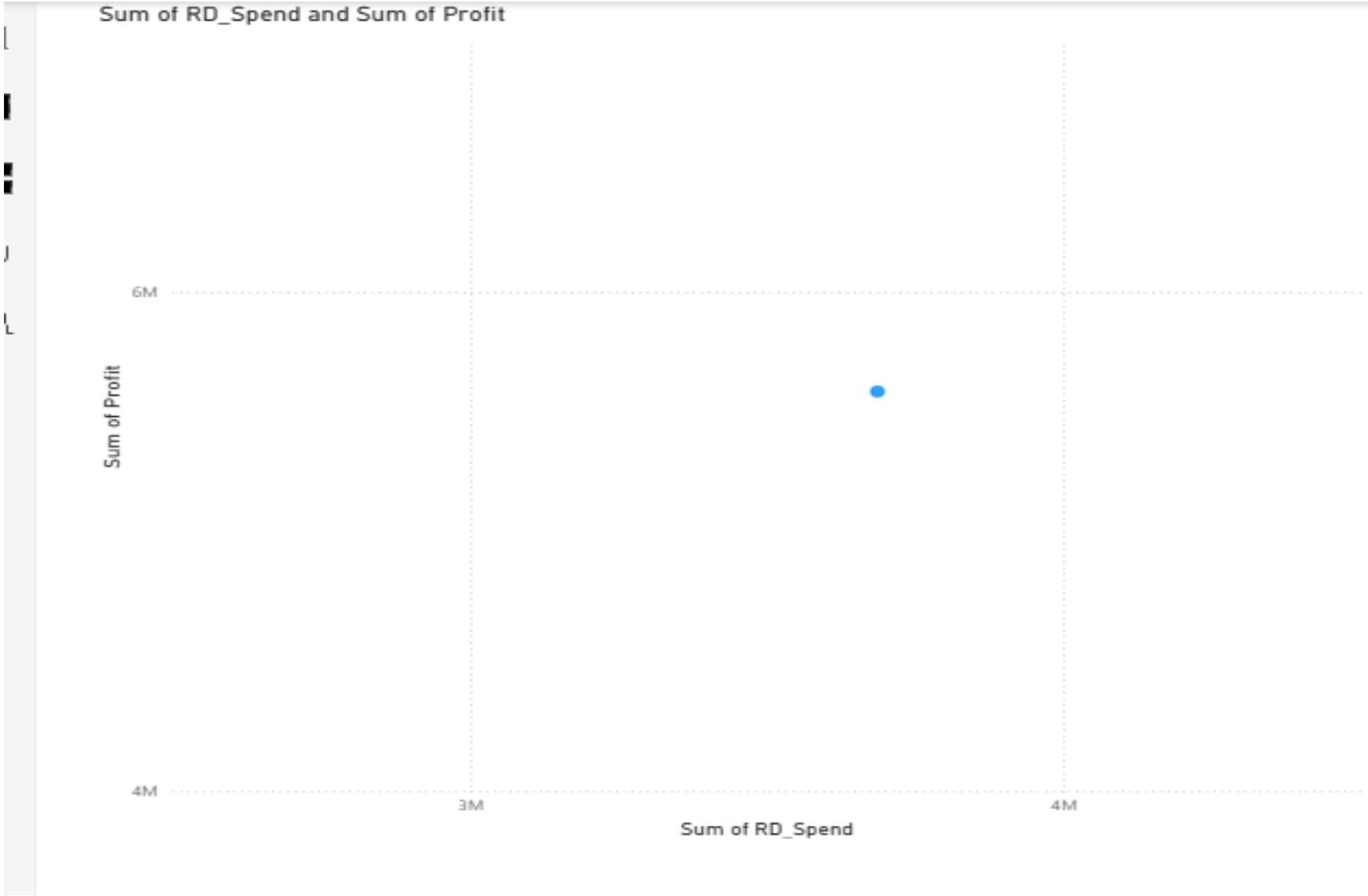
Cont..

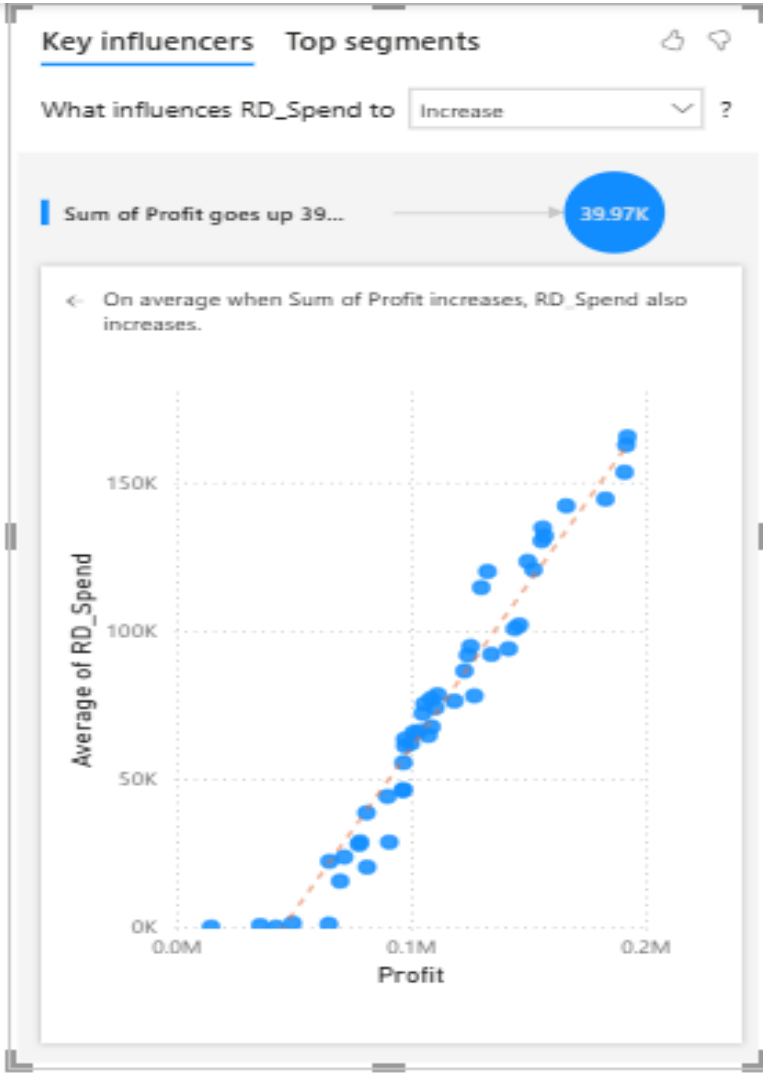
- **The Small Circle (Far Left): This is your Outlier. This represents one specific startup that made significantly less profit (around 15,000 to 20,000) compared to everyone else.**

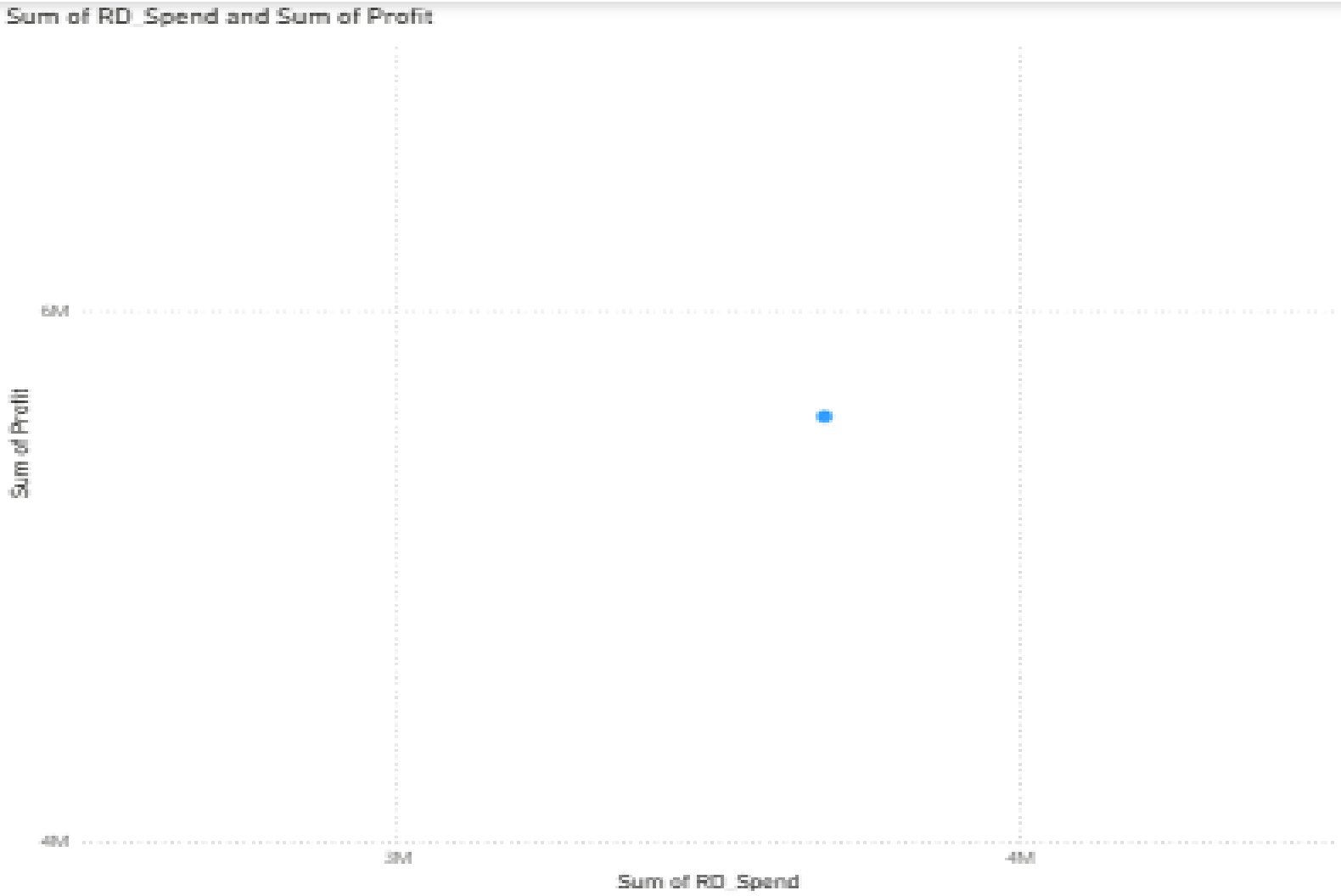
Cont…

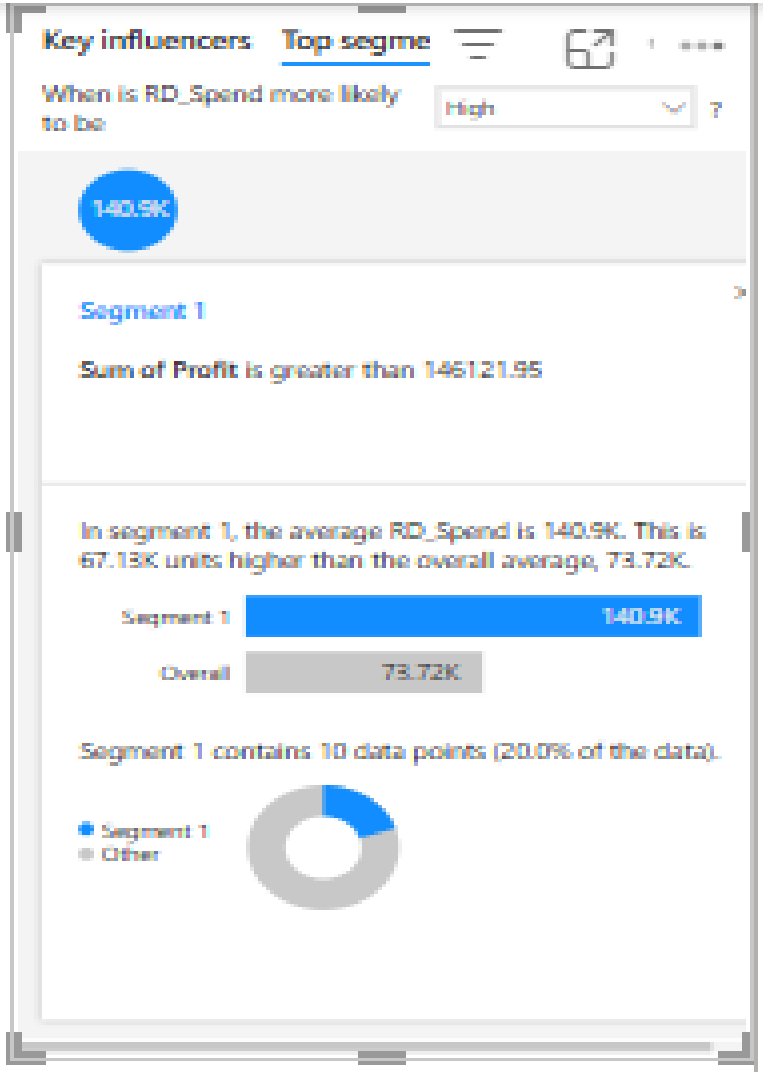
- **Outliers are "danger zones" for Regression.**
  - ❖**Because that one startup has such low profit, it might "pull" your regression line down, making your prediction less accurate.**

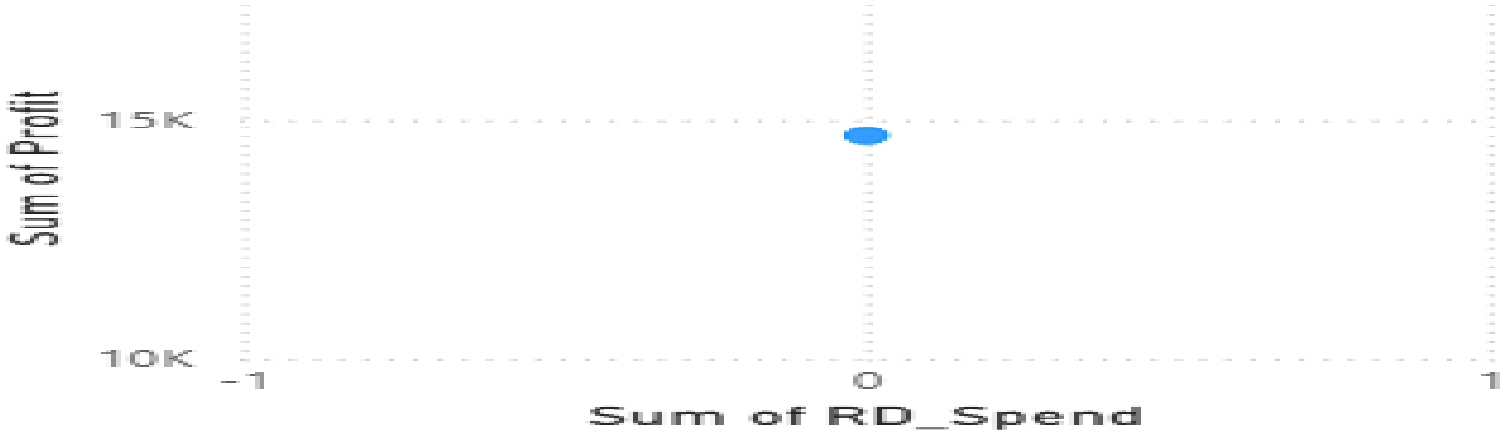
Cont..


Sum of RD_Spend and Sum of Profit
6M
4M
Sum of Profit
3M
4M
Sum of RD_Spend


Key influencers
Top segments
What influences RD_Spend to
Increase
?
Sum of Profit goes up 39...
39.97K
On average when Sum of Profit increases, RD_Spend also increases.
150K
100K
50K
0K
Average of RD_Spend
0.0M
0.1M
0.2M
Profit

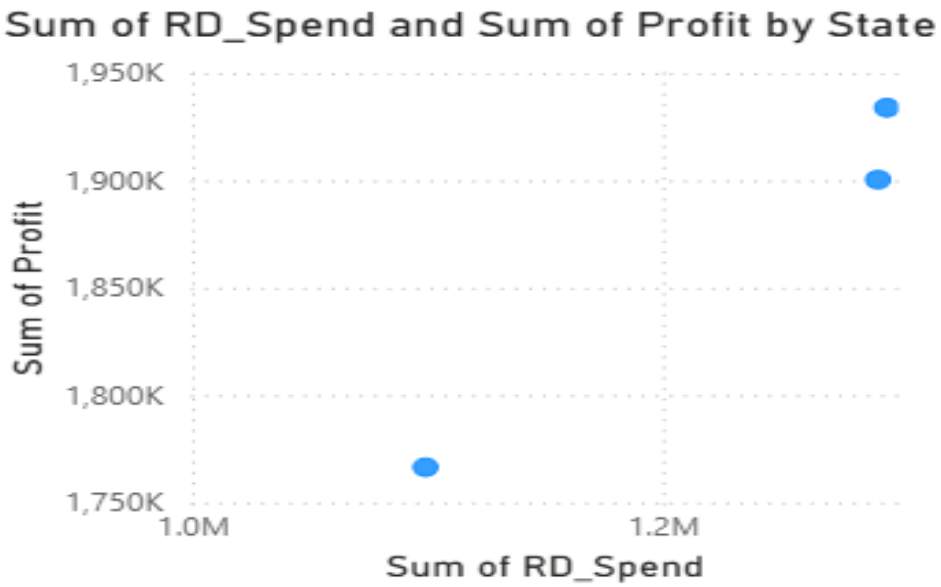

Sum of RD_Spend and Sum of Profit
6M
4M
Sum of Profit
3M
4M
Sum of RD_Spend


Key influencers
Top segme
When is RD_Spend more likely to be
High
?
140.9K
Segment 1
Sum of Profit is greater than 146121.95
In segment 1, the average RD_Spend is 140.9K. This is 67.13K units higher than the overall average, 73.72K.
Segment 1
140.9K
Overall
73.72K
Segment 1 contains 10 data points (20.0% of the data).
Segment 1
Other



Outlier Label	Sum of Profit	Sum of RD_Spend
Normal	5,585,950.56	3,686,080.78
Outlier	14,681.40	0.00
Total	**5,600,631.96**	**3,686,080.78**

Cont…



State	Sum of RD_Spend	Sum of Profit
California	1,099,180.46	1,766,387.98
Florida	1,291,584.26	1,900,384.39
New York	1,295,316.06	1,933,859.59
Total	**3,686,080.78**	**5,600,631.96**

Outlier Label	Sum of Profit	Sum of RD_Spend	State
Normal	1,933,859.59	1,295,316.06	New Y
Normal	1,900,384.39	1,291,584.26	Florida
Normal	1,751,706.58	1,099,180.46	Califor
Outlier	14,681.40	0.00	Califor
Total	**5,600,631.96**	**3,686,080.78**	

Summary for the Outlier Analysis

- Observation: The analysis identified a significant outlier in the dataset located in California.
- The Data: While most profitable startups in New York, Florida, and California invest heavily in R&D (averaging over 1M$), one specific entry recorded a profit of 14,681.40$ despite having 0.00$ R&D Spend.

2. Visual Interpretation (Scatter Chart)

❖**Graph Title: Sum of RD_Spend and Sum of Profit by State**

- ✓**The Trend: There is a clear Positive Correlation between R&D Spend and Profit. As R&D Spend increases (moving right on the X-axis), Profit increases (moving up on the Y-axis).**
- ✓**The Cluster: New York and Florida startups are clustered at the top-right, showing high investment and high returns.**
- ✓**The Deviation: The outlier (California) sits at the far left of the chart. This represents a "Bootstrap" business model—generating profit without the traditional R&D costs associated with tech startups.**

The Concerns

- **Technical Risk: Feature Skewness**
- **The presence of a 0.00$ R&D value alongside a positive Profit creates a "Skew" in the Linear Regression model. If this outlier is included in the training set, the model might underestimate the importance of R&D for the other 49 startups.**
- **Action Taken: I implemented a DAX-based Outlier Labeling system to segment this data. This allows me to run two scenarios:**

Cont..

- **Action Taken: I implemented a DAX-based Outlier Labeling system to segment this data. This allows me to run two scenarios:**
  1. **Full Dataset: Includes all startups (Lower accuracy).**
  2. **Cleaned Dataset: Excludes the 0$ R&D outlier (Higher predictive precision).**

Strategic Recommendations

Based on the regression analysis and the outlier detection in Power BI, I propose the following three actions for the company:

A. Prioritize R&D in New York and Florida

The data shows the most stable and predictable "Return on Investment" (ROI) in these states. Every dollar spent on R&D here consistently moves the startup up the profit line.

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- **Action: Increase R&D budget by 10% in these regions to maximize high-confidence profit growth.**

B. Investigate the "California Anomaly“

The outlier I identified (Zero R&D but $14k Profit) suggests there is a specific "Service-Only" or "Low-Tech" business model working in California.

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- **Action: Conduct a qualitative interview with the California team to see if their "No-R&D" model can be scaled to other states to save costs.**

C. Standardize Data Entry Protocols

- **The difficulty in matching column names (e.g., RD_Spend vs R&D Spend) and the discovery of zero-values indicates a need for better data governance.**

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- **Action: Implement a standardized data entry template in the MySQL database to ensure future regression models don't require extensive manual cleaning.**

Conclusion & Recommendations

- ✓ Focus on Innovation: Increasing R&D budget is the fastest way to scale profit.
- ✓ Optimize Marketing: Marketing spend is effective, but ROI(Return on Investment) should be tracked closely.
- ✓ Control Admin: Keep administrative costs lean as they do not contribute to profit margins.

Final Project Conclusion

- "Through the integration of Python-based Linear Regression and Power BI Visual Analytics, this project successfully identified the key drivers of startup profitability. By isolating outliers and validating the positive correlation between R&D and Profit, I have created a predictive model that can forecast future earnings with high reliability. This data-driven approach minimizes the risk of 'Guesswork' in budget allocation."

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